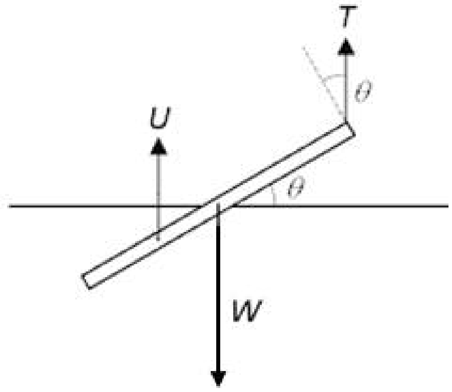
7	**(a)**	**(i)**	The rod is in vertical equilibrium only if Tension is acting vertically upwards, since Upthrust and Weight are also acting vertically.	
		(ii)	 Where W is the weight of the rod and T is the tension on the string U is the upthrust on the rod	
		(iii)	Taking moments about the centre of gravity of the rod, $U\cos\theta\left(\frac{0.5}{2}\right)=T\cos\theta(0.5)$ $0.5\ U=T$	
		(iv)	Balancing vertical forces, $T + U = W$ Using, $0.5\ U = T$ We have, $T + 2T = W$ $T = 16.7\ N$	
		(v)	Balancing vertical forces, $T + U = W$ $0.5\ U + U = W$ $U=\frac{2}{3}W$ $U= V_f\rho_f g$ Since volume of the rod, V_r is equal to the twice the volume of the fluid displaced, $2V_f$ $U= 0.5V_r\rho_f g$ $\frac{2U}{\rho_f g}=V_r$ We have, $W= V_r\rho_r g$ $W= \frac{2U}{\rho_f g}\rho_r g$ $W= \frac{4W}{3\rho_f g}\rho_r g$	

			$$\rho_r = \frac{3(1000)}{4} = 750\ kg\ m^{-3}$$	
	(b)	**(i)**	The total linear momentum of a system will remain constant if no net external force acts on it.	
		(ii)	Taking vectors to the right as positive. Using conservation of linear momentum $1.2u+0.6(-0.2)=1.2v+0.6(0.1)$ $1.2u=1.2v+0.18$----(1) Using relative speed of approach = relative speed of separation: $u+0.2=0.1-v$-----(2) Solving (1) and (2) u = 0.025 m s^{-1}	
	(c)	**(i)**	When the gas is ejected from the thruster, it gains a momentum. As a consequence, based on the conservation of momentum, the astronaut gains an equal and opposite momentum. Hence the gas is to be ejected in the opposite direction to that in the direction of the space shuttle. **OR** The thruster exerts a force onto the gas. By Newton's third law, the gas exerts an equal and opposite force on the thruster and the astronaut. To allow the astronaut to move towards the space shuttle, the gas is to be ejected in the opposite direction to that in the direction of the space shuttle.	
		(ii)	Based on the conservation of momentum, $m\Delta u = M\Delta v$ (0.5)(1)(20 – 0) = 80 (v – 0) v = 0.125 m s^{-1} OR Magnitude of momentum of gas $F\Delta t = \Delta p$ $\frac{dp}{dt}\Delta t = m\Delta v$ $(0.5)(20)(1.0)=80(v-0)$ $v = 0.125\ m\ s^{-1}$	

8	**(a)**	**(i)**	It is the energy transferred per unit charge from other forms into electrical energy when	